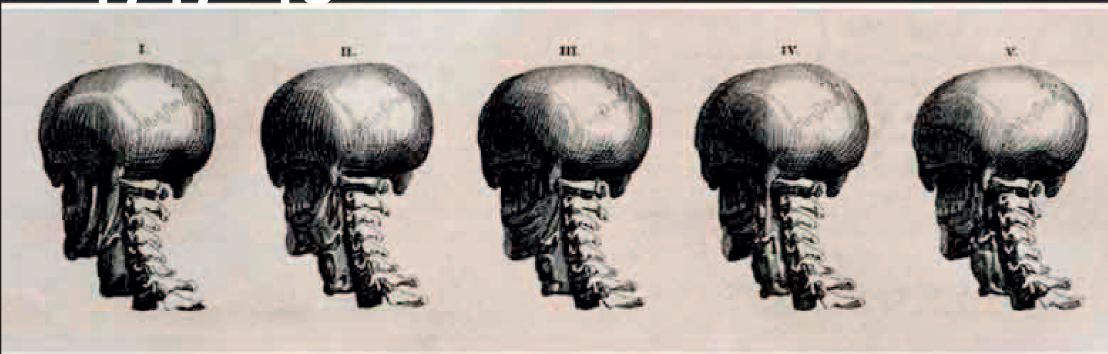
1747–48



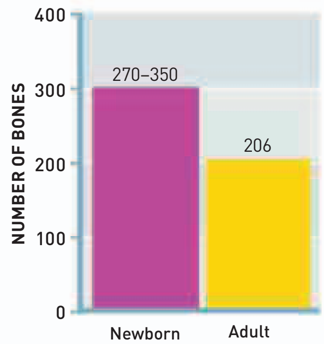
Bernhard Siegfried Albinus's *Tabulae Sceleti et Musculorum Corporis Humani* contained anatomical drawings of unprecedented accuracy.

AFTER ENGLISH PHYSICIST ISAAC NEWTON'S DISCOVERY of the law of gravitation (see 1687–89) many others became interested in the **gravitational effect of the Moon**. In 1747, French mathematician **Jean le Rond d'Alembert** argued that winds are caused by "tides" in the atmosphere that are driven by the Moon, just like tides in the sea. He was mistaken—winds are really driven by variations in the way the air is warmed by the sun, as warm air rises and cold air rushes in to take its place. However, his work did introduce **partial differential equations (PDEs)**, complex equations involving several variables. Later developed by Swiss mathematician **Leonhard Euler**, PDEs are now used for **calculating how fast one variable changes** when others are held constant, and they are central to calculations about the movement of sound, heat, electricity, and fluids.

Countless sailors were **dying from scurvy** on long voyages. Nobody knew at the time that the illness was caused by a **vitamin C deficiency**, but some people suspected it might be prevented by eating lemons and limes. In 1747, British naval surgeon **James Lind** (1716–94) carried out an experiment to test the effect of different dietary supplements on six pairs of sailors suffering from scurvy. Only the pair fed limes recovered, and we now know that eating **citrus fruit prevents scurvy** because it contains vitamin C.

In 1748, Dutch anatomist **Bernhard Siegfried Albinus** (1697–1770) published an important **study of human anatomy**, entitled *Tabulae Sceleti et Musculorum Corporis Humani* (*Drawings of the Skeleton and Muscles of the Human Body*). The drawings were plotted on grids to ensure their accuracy.

Also in 1748, English physicist **James Bradley** explained an astronomical effect he had been studying for 20 years. This was **nutation**—the way Earth's axis nods slightly with a period of 18.6 years. The Moon's orbit does not lie exactly in the plane of the ecliptic (Earth's orbit around the Sun), so its changing unsymmetrical gravitational pull unbalances Earth's rotation.



Human bones
Babies have more bones than adults, and some adult bones result from the fusion of bones that are separate in newborns.

Treating scurvy
James Lind's 1747 study proved that citrus fruits prevented scurvy, but it was many years before his ideas were put into practice.

1747 Jean le Rond d'Alembert develops **partial differential equations**

1747 James Lind suggests that **citrus fruit cures scurvy**

1748 English physician **John Fothergill** describes the illness **diphtheria**

1748 James Bradley publishes his explanation of **Earth's nutation**

1748 Bernhard Siegfried Albinus publishes his study of human anatomy

1749–50

English astronomer Thomas Wright described the Milky Way as being shaped like a disk.

IN 1749, FRENCH NATURALIST GEORGES BUFFON (1707–88) began the publication of his study *Histoire Naturelle* (*Natural History*), a 44-volume **study of animals and minerals**. He was one of the first to recognize that the world is very ancient and that many species have come and gone since it was formed. This laid the groundwork for Darwin's theory of evolution a century later (see pp.204–205).

Also this year, Swiss mathematician **Leonhard Euler** turned his attention to the stability of ships in his book *Scientia Navalis* (*Naval Science*). He analysed their **three-dimensional motion** at sea with such mathematical precision that he had to add a third axis to graphs to show

Buffon's *Natural History*
This turkey is one of the many accurately observed drawings in Georges Buffon's important study, Histoire Naturelle*, a work translated into several languages.*

1749 Leonhard Euler introduces **3D coordinates**

April 12, 1749 Leonhard Euler proves **Pierre de Fermat's theorem for prime numbers**

1749 Pierre Bouguer publishes his expedition findings in *La figure de la Terre*

1749 Georges de Buffon begins publication of his study of natural history, *Histoire Naturelle*